

GI IMAGING (Radiology)

PLAIN ABDOMINAL X-RAY

- Standard = **supine AP**. **Free intraperitoneal air best on ERECT CHEST X-ray**. Can't sit → **lateral decubitus**.
- **Gas pattern**: large amounts normal in stomach + colon. Stomach = LUQ (gastric rugae bands). Small bowel = little gas.

SBO vs LBO (HIGH-YIELD)

| Feature | Small bowel obstruction | Large bowel obstruction |

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| Location | **Central** ("frame" of colon) | **Peripheral** (transverse/sigmoid central) |

| Loops | Numerous | Few |

| Mucosal markings | **Valvulae conniventes** (cross full width, "stack of coins") | **Haustra** (incomplete bands, absent distal to splenic flexure) |

| Curve | Tight | Wide |

| Diameter | **3–6 cm** | **>6 cm** |

- **Mechanical obstruction causes**: lumen (gallstone ileus), wall (tumour, Crohn's), extrinsic (adhesions, hernia). Non-mechanical dilatation: paralytic ileus, ischaemia, IBD.
- **CT**: shows **transition point** (dilated→collapsed), confirms/excludes mass; **small bowel feces sign**.

Pneumoperitoneum

- Commonest spontaneous cause = **perforated peptic ulcer** (2/3 recognizable radiologically). Normal post-laparotomy (absorbs <7 days adults, <24h children). **Erect CXR: curvilinear gas under right hemidiaphragm** (between diaphragm + liver).

Abdominal calcification

- **Appendicolith** + appendicitis features → strong indication (often gangrene/perforation). **Phleboliths** (pelvic veins, mistaken for stones). Calcified mesenteric LN (old TB, mobile). **Vascular = curvilinear (AAA wall)**. Uterine fibroids. **Dermoid cyst = tooth shape**. Adrenal (haemorrhage/TB/Addison's minority). **Gallstones 20% visible (dense rim, lucent centre)**. **Pancreatic calcification = chronic pancreatitis (L1 level)**. Renal **staghorn stones**.

GI IMAGING MODALITIES

- **Barium sulphate**: best contrast/opacification. **Contraindicated if perforation suspected** (causes severe peritonitis if leak).
- **Gastrografin (water-soluble)**: for **suspected perforation/anastomotic leak**; less radio-opaque; **irritant to lungs** → **pulmonary oedema** if aspirated.
- **Double-contrast** (barium + air): mucosal detail (stomach, colon).
- **Ultrasound**: fluid, bowel wall; **pyloric stenosis, intussusception, appendicitis, diverticulitis**.
- **EUS: depth of tumour invasion** (esophagus, gastric, rectal); small pancreatic/duodenal tumours; guide FNA/biopsy.
- **CT**: full GI wall + surrounding fat (not mucosa); staging tumours, inflammation, obstruction, trauma. Mesenteric CTA for ischaemia.
- **MRI**: limited (peristalsis artefact) but **rectal carcinoma staging + MR enterography (small bowel/Crohn's, antiperistaltic agent)**.
- **FDG-PET/CT**: metastatic disease in GI cancers.

Descriptive terms (HIGH-YIELD)

- **Ulcer:** breach of mucosa, barium fills crater (projects beyond wall).
- **Filling defect:** intraluminal (food, surrounded by barium), intramural (tumour/leiomyoma, indents), extramural (compresses, mucosa preserved).
- **Stricture:** circumferential narrowing. **Benign = tapering ends; malignant = abrupt with overhanging edges ("shouldering").**
- **Fat stranding** (CT/MRI) / echogenic fat (US) = inflammation. Wall thickening. FDG uptake (PET).

ESOPHAGUS

- Normal: smooth, indentations from aortic arch + left atrium. **Tertiary contractions = corkscrew oesophagus.**
- **Strictures:**
- **Carcinoma:** irregular lumen + **shouldered edges**, several cm; CT/EUS/PET for staging.
- **Peptic:** lower end, **short, smooth, tapering.**
- **Achalasia:** failure of cardiac sphincter relaxation → **smooth tapered narrowing at lower end + dilated oesophagus above with absent peristalsis** ("bird's beak"); plain film = food residue, aspiration → consolidation/bronchiectasis.
- **Corrosive:** long stricture from aortic arch level; smooth/tapered ± irregular.
- **Leiomyoma** (benign): smooth rounded indentation.
- **Dilatation:** with obstruction (cancer/achalasia); **without obstruction = scleroderma** (lower 2/3 smooth muscle).
- **Zenker's diverticulum:** posterior pharynx (inferior constrictor weakness). **Web:** cervical, anterior, **Plummer-Vinson.**
- **Perforation:** X-ray pneumomediastinum + effusion; **water-soluble contrast** shows leak; **CT with oral contrast most sensitive.**

STOMACH & DUODENUM

- **Peptic ulcer:** gastric = benign or malignant (**always scope**); duodenal = almost always benign. Ulcer = barium projection beyond mucosa; duodenal cap deformed by scarring.

Benign vs malignant gastric ulcer (HIGH-YIELD)

| Benign | Malignant |

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| Crater **projects beyond** gastric contour | Does NOT protrude beyond contour (within outline) |

| Smooth, rounded, deep crater | Irregular, shallow crater |

| Smooth folds **reaching** ulcer margin | Nodular folds **NOT reaching** margin |

| **Hampton's line** | **Carman meniscus sign** |

- **Gastric carcinoma:** irregular filling defect; or focal wall thickening; or **linitis plastica** (diffuse thickening, rigid stomach, loss of folds). Staging: CT/EUS/laparoscopy.
- **GIST:** smooth round submucosal filling defect; usually benign; **60–70% in stomach.**
- **Lymphoma:** stomach = most frequent GI site; diffuse wall thickening + bulky lymphadenopathy.
- **Hiatus hernia:** **sliding** (GEJ + stomach above diaphragm, incompetent sphincter → reflux/stricture); **rolling/paraesophageal** (fundus herniates, GEJ competent). CXR = retrocardiac mass + air-fluid level.

SMALL & LARGE INTESTINE

Techniques

- **Small bowel follow-through** (drink barium): Crohn's, stricture, malabsorption. **Enteroclysis** (NJ tube + barium + methylcellulose): best mucosal detail. **Barium enema** (double contrast): colon.
- CT, US (bowel thickening, appendix, intussusception), MRI (rectal/anal, MR enterography). **Tc-99m for Meckel's**; octreotide/MIBG for carcinoid.

Dilatation/narrowing

- Small >3cm (malabsorption/ileus/SBO); large >6cm (obstruction, ileus, volvulus, **toxic megacolon**, Hirschsprung).
- **Colonic stricture site**: diverticular = **sigmoid**; ischaemic = **splenic flexure–sigmoid**; Crohn's/TB = **caecum**. Neoplastic = shouldered, irregular, <6cm; benign = tapered.

IBD (HIGH-YIELD)

| Feature | Ulcerative colitis | Crohn's |

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| Distribution | **Continuous from rectum** | **Skip lesions**, terminal ileum |

| Ulcers | Shallow (deep if severe) | Deep → **cobblestone** |

| Haustra | **Lost** → rigid tube ("lead pipe") | — |

| Other | **Pseudopolyps**, backwash ileitis | **String sign**, fistulae, bowel separation, caecal contraction |

| CT/MRE | Wall thickening, "**target sign**" (fatty submucosa) | Thick wall + **fat stranding**, prominent mesenteric vessels |

| Complication | **Toxic megacolon**; stricture rare (→ think cancer) | Strictures, fistulae, abscess |

Diverticular disease

- Sac-like mucosal outpouchings through muscle; commonest **sigmoid**. **Diverticulitis** = infected (thick wall + fat stranding ± perforation/abscess). Stricture **indistinguishable from carcinoma**.

Appendicitis

- Clinical mostly. US/CT: **distended >6mm, non-compressible appendix, thick wall, free fluid, fat inflammation**; abscess = RIF mass.

Obstruction & emergencies

- **SBO = adhesions/hernia; LBO = tumours**. CT transition zone; adhesions can't be directly imaged.
- **Volvulus**: bowel twists on mesentery; **sigmoid commonest** (then caecum). **Coffee-bean sign** (AXR); **whirl sign** (CT, twisted vessels).
- **Intussusception**: invagination. **Children = idiopathic** (US target/doughnut, air/CO2 enema reduction); **adults = neoplasm lead point** (CT sausage-shaped mass with mesenteric fat).
- **Mesenteric infarction**: SMA occlusion → wall thickening, **gas in bowel wall**; CT angiography shows thrombus. **Ischaemic colitis**: splenic flexure–sigmoid; **thumbprint sign** (mucosal oedema/haemorrhage).
- **TB**: contracted caecum, dilated terminal ileum, peritoneal nodules + necrotic nodes — mimics Crohn's/malignancy → biopsy.

Polyps & tumours

- **Polyp malignancy risk**: >2cm, short thick stalk, irregular surface. <1cm rarely cancer.
- **Carcinoma**: commonest **rectosigmoid + caecum**. Rectosigmoid → annular stricture + obstruction; **caecal** → **large without obstruction (anaemia + weight loss)**. Barium = **shouldered irregular stricture <6cm** or fungating mass. **MRI for rectal staging**; CT for mets; PET before curative surgery.

- **Lymphoma:** marked wall thickening **WITHOUT lumen narrowing** (differentiates from Crohn's); mesenteric/para-aortic nodes.

HIGH-YIELD ONE-LINERS

- **SBO** = central, valvulae conniventes (stack of coins), 3–6cm; **LBO** = peripheral, haustra, >6cm.
- **Free air** = erect CXR (perforated PU commonest). Barium contraindicated if perforation → use Gastrografin.
- **Benign ulcer projects beyond contour** (Hampton's line); **malignant within contour** (Carman sign).
- **Achalasia** = bird's beak; **scleroderma** = dilated no obstruction; **corrosive** = long from aortic arch.
- **UC** = continuous, lost haustra (lead pipe), pseudopolyps, toxic megacolon; **Crohn's** = skip lesions, cobblestone, string sign, fistulae.
- **Volvulus** = coffee-bean / whirl sign; sigmoid commonest.
- **Intussusception:** children idiopathic (reduce by enema); adults = tumour lead point.
- **Ischaemic colitis** = thumbprint sign, splenic flexure–sigmoid.
- **Lymphoma** = thick wall, NO narrowing (vs Crohn's). Polyp malignant if >2cm.
- **Linitis plastica** = rigid stomach; **GIST** = submucosal; gastric lymphoma commonest GI lymphoma site.